

GOLANG MOSES SEKGOGOBA

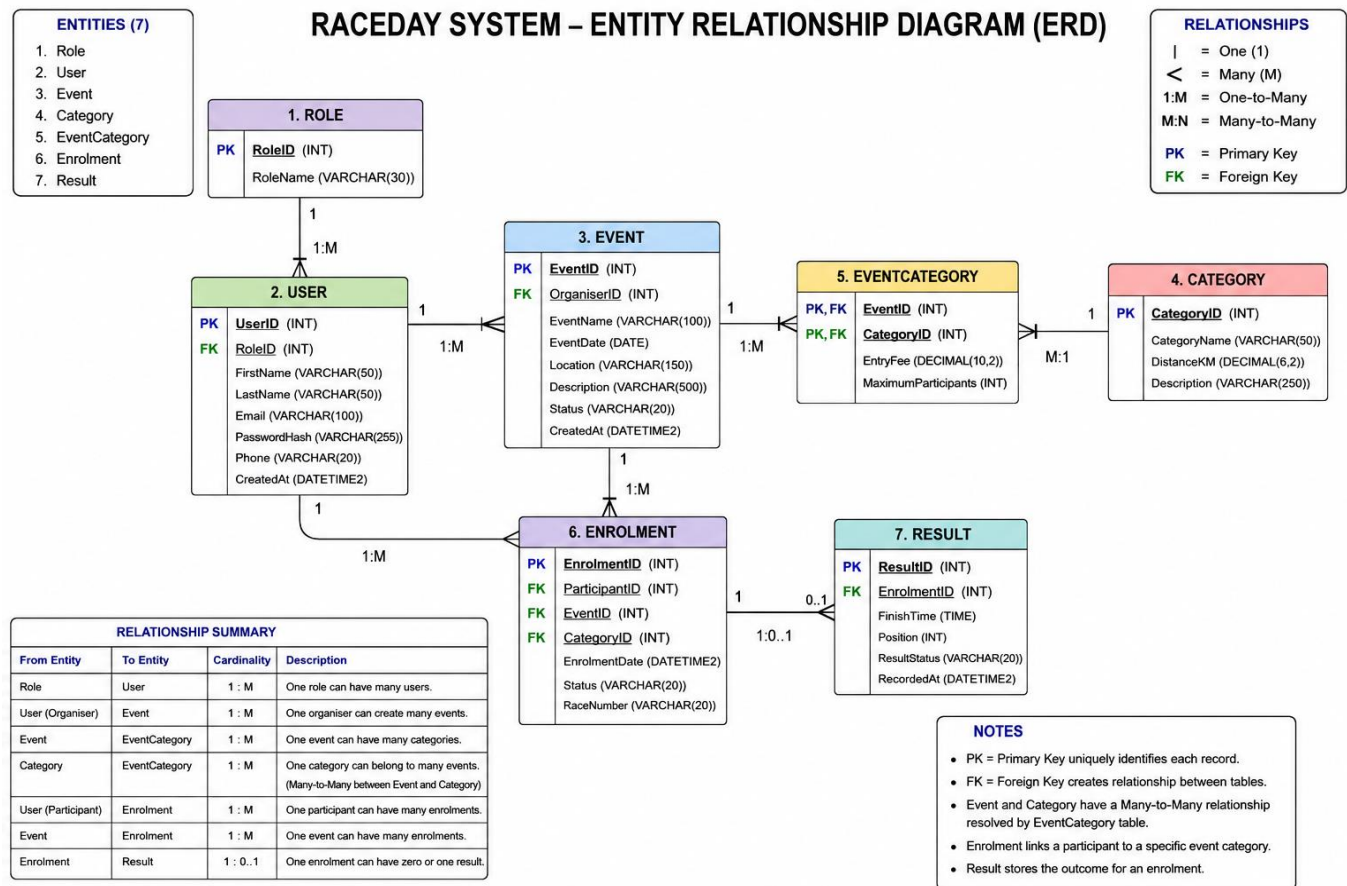
ST10480178

GROUP 5 (PROG 6212)

ASSIENMENT PART 1

QUESTION1

SECTION A



SECTION B

1. Authentication Endpoints

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
POST	/api/auth/register	Registers a new user on the RaceDay system.	None	{ firstName, lastName, email, password, roleId, phone }	201 Created – User registered successfully. 400 Bad Request – Invalid details.
POST	/api/auth/login	Authenticates a user and creates an access token.	None	{ email, password }	200 OK – Login successful and token returned. 401 Unauthorized – Invalid credentials.
GET	/api/auth/me	Retrieves the details of the currently logged-in user.	Logged-in user	None	200 OK – Current user details. 401 Unauthorized – Not authenticated.

2. User Profile Endpoints

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/users/{id}	Retrieves a user's profile using their UserID.	Logged-in user/Admin	None	200 OK – User profile returned. 404 Not

					Found – User does not exist.
PUT	/api/users/{id}	Updates a user's profile information.	Account owner/Admin	{ firstName, lastName, phone }	200 OK – Profile updated. 404 Not Found – User does not exist.

3. Event Endpoints

These endpoints manage the Event table from our ERD.

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/events	Retrieves all RaceDay events.	Public	None	200 OK – List of events returned.
GET	/api/events/{id}	Retrieves a specific event using EventID.	Public	None	200 OK – Event details returned. 404 Not Found – Event does not exist.
POST	/api/events	Creates a new RaceDay event.	Organiser	{ eventName, eventDate, location, description }	201 Created – Event created successfully. 400 Bad Request – Invalid event data.
PUT	/api/events/{id}	Updates an existing RaceDay event.	Organiser	{ eventName, eventDate, location, description, status }	200 OK – Event updated. 404 Not Found –

					Event does not exist.
DELETE	/api/events/{id}	Deletes an existing event.	Organiser	None	204 No Content – Event deleted. 404 Not Found – Event does not exist.

4. Category Endpoints

These endpoints manage race categories such as **5 KM, 10 KM, 21 KM and 42 KM**.

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/categories	Retrieves all available race categories.	Public	None	200 OK – List of categories returned.
GET	/api/categories/{id}	Retrieves a specific category.	Public	None	200 OK – Category details. 404 Not Found – Category does not exist.
POST	/api/categories	Creates a new race category.	Administrator	{ categoryName, distanceKM, description }	201 Created – Category created. 400 Bad Request – Invalid data.
PUT	/api/categories/{id}	Updates an existing	Administrator	{ categoryName,	200 OK – Category

		race category.		distanceKM, description }	updated. 404 Not Found – Category does not exist.
DELETE	/api/categories/{id}	Deletes a race category.	Administrator	None	204 No Content – Category deleted.

5. Event Category Endpoints

The EventCategory table connects **Events** and **Categories**.

Remember:

One event can have many categories, and one category can be used by many events.

Therefore, this table resolves the **many-to-many relationship**.

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/events/{eventId}/categories	Retrieves all categories available for a specific event.	Public	None	200 OK – Event categories returned.
POST	/api/events/{eventId}/categories	Adds a category to an event.	Organiser	{ categoryId, entryFee, maximumParticipants }	201 Created – Category added to event.

					409 Conflict – Category already assigned.
PUT	/api/events/{eventId}/categories/{categoryId}	Update the fee or participant limit for an event category.	Organiser	{ entryFee, maximumParticipants }	200 OK – Event category updated.
DELETE	/api/events/{eventId}/categories/{categoryId}	Removes a category from an event.	Organiser	None	204 No Content – Category removed.

6. Event Enrolment Endpoints

These endpoints allow participants to enter races.

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
POST	/api/events/{eventId}/enrolments	Registers the logged-in participant for an event	Participant	{ categoryId }	201 Created – Enrolment created. 409

		category.			Conflict – Participant already enrolled.
GET	/api/enrolments/me	Retrieves all events in which the logged-in participant is enrolled.	Participant	None	200 OK – Participant enrolments returned.
GET	/api/enrolments/{id}	Retrieves a specific enrolment.	Participant/Organiser	None	200 OK – Enrolment details. 404 Not Found – Enrolment not found.
GET	/api/events/{eventId}/enrolments	Retrieves all participants enrolled in an event.	Organiser	None	200 OK – List of participants.
PUT	/api/enrolments/{id}	Updates the status of an enrolment.	Participant/Organiser	{ status }	200 OK – Enrolment updated.
DELETE	/api/enrolments/{id}	Cancels a	Participant/Organiser	None	204 No Content

		participant's enrolment.			– Enrolment cancelled.
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7. Results Endpoints

These endpoints manage participant race results.

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
POST	/api/enrolments/{enrolmentId}/result	Records the result of a participant who completed a race.	Organiser	{ finishTime, position, resultStatus }	201 Created – Result recorded. 404 Not Found – Enrolment not found.
GET	/api/enrolments/{enrolmentId}/result	Retrieves the result associated with an enrolment.	Participant/Organiser	None	200 OK – Result returned. 404 Not Found – Result not available.
GET	/api/events/{eventId}/results	Retrieves all	Public	None	200 OK –

		results for a particular event.			Event results returned.
GET	/api/results/{id}	Retrieves a specific race result.	Public	None	200 OK – Result details. 404 Not Found – Result not found.
PUT	/api/results/{id}	Updates an existing race result.	Organiser	{ finishTime, position, resultStatus }	200 OK – Result updated. 404 Not Found – Result not found.
DELETE	/api/results/{id}	Removes an incorrect race result.	Organiser	None	204 No Content – Result deleted.

8. Role Endpoints

Because our database has a Role table, we can include these endpoints as well.

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
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GET	/api/roles	Retrieves all roles available in the system.	Administrator	None	200 OK – List of roles returned.
GET	/api/roles/{id}	Retrieves a specific role.	Administrator	None	200 OK – Role details returned. 404 Not Found – Role not found

SECTION C

FULL CODE & RESULT

SQL Query 2 (71) * SQL Query 1 (52)

```
SELECT * FROM dbo.Role;  
SELECT * FROM dbo.[User];  
SELECT * FROM dbo.Event;  
SELECT * FROM dbo.Category;  
SELECT * FROM dbo.EventCategory;  
SELECT * FROM dbo.Enrolment;  
SELECT * FROM dbo.Result;  
GO
```

Results Messages

RoleID	RoleName
1	Administrator
2	Organiser
3	Participant

UserID	RoleID	FirstName	LastName	Email	PasswordHash	Phone	CreatedAt
1	1	John	Mokoena	john.mokoena@raceday.co.za	HASHED_PASSWORD_001	0711111111	2026-09-07 12:28:21.7966667
2	2	Maria	Molefe	maria.molefe@raceday.co.za	HASHED_PASSWORD_002	0722222222	2026-09-07 12:28:21.7966667
3	3	Thabo	Nkosi	thabo.nkosi@gmail.com	HASHED_PASSWORD_003	0733333333	2026-09-07 12:28:21.7966667
4	2	Lerato	Mabena	lerato.mabena@gmail.com	HASHED_PASSWORD_004	0744444444	2026-09-07 12:28:21.7966667
5	3	Admin	RaceDay	admin@raceday.co.za	HASHED_PASSWORD_005	0755555555	2026-09-07 12:28:21.7966667

EventID	OrganiserID	EventName	EventDate	Location	Description	Status	CreatedAt
1	1	Polokwane City Marathon	2026-10-10	Polokwane	Annual road running event in Polokwane.	Open	2026-09-07 12:28:21.7966667
2	2	Limopo Fun Run	2026-11-07	Tzaneen	Community fun run for families and runners.	Open	2026-09-07 12:28:21.7966667
3	3	Spring Cycling Race	2026-09-20	Mokopane	Competitive cycling race for different cat...	Open	2026-09-07 12:28:21.7966667

Query executed successfully. LabVM2049939\SQLEXPRESS (17... LabVM2049939\Student (71) RaceDayDB | 00:00:01 Row: 1, Col: 1 48 rows

Error List

Search Error List

OUT PUT

RoleID	RoleName
1	Administrator
2	Organiser
3	Participant

UserID	RoleID	FirstName	LastName	Email	PasswordHash	Phone	CreatedAt
1	1	John	Mokoena	john.mokoena@raceday.co.za	HASHED_PASSWORD_001	0711111111	2026-09-07 12:28:21.7966667
2	2	Maria	Molefe	maria.molefe@raceday.co.za	HASHED_PASSWORD_002	0722222222	2026-09-07 12:28:21.7966667
3	3	Thabo	Nkosi	thabo.nkosi@gmail.com	HASHED_PASSWORD_003	0733333333	2026-09-07 12:28:21.7966667
4	2	Lerato	Mabena	lerato.mabena@gmail.com	HASHED_PASSWORD_004	0744444444	2026-09-07 12:28:21.7966667
5	3	Admin	RaceDay	admin@raceday.co.za	HASHED_PASSWORD_005	0755555555	2026-09-07 12:28:21.7966667

EventID	OrganiserID	EventName	EventDate	Location	Description	Status	CreatedAt
1	1	Polokwane City Marathon	2026-10-10	Polokwane	Annual road running event in Polokwane.	Open	2026-09-07 12:28:21.7966667
2	2	Limopo Fun Run	2026-11-07	Tzaneen	Community fun run for families and runners.	Open	2026-09-07 12:28:21.7966667
3	3	Spring Cycling Race	2026-09-20	Mokopane	Competitive cycling race for different cat...	Open	2026-09-07 12:28:21.7966667

CategoryID	CategoryName	DistanceKM	Description
1	5 KM	5.00	Short distance race
2	10 KM	10.00	Intermediate distance race
3	21 KM	21.10	Half marathon distance
4	42 KM	42.20	Full marathon distance

OUTPUT

EventID	CategoryID	EntryFee	MaximumParticipants
1	1	100.00	500
2	1	150.00	500
3	1	250.00	300
4	1	350.00	200
5	2	50.00	300
6	2	100.00	250
7	3	200.00	200
8	3	300.00	150

EnrolmentID	ParticipantID	EventID	CategoryID	EnrolmentDate	Status	RaceNumber
1	1	3	1	2026-09-07 12:28:21.8333333	Confirmed	R001
2	2	4	1	2026-09-07 12:28:21.8333333	Confirmed	R002
3	3	3	2	2026-09-07 12:28:21.8333333	Confirmed	R003
4	4	4	3	2026-09-07 12:28:21.8333333	Confirmed	R004

ResultID	EnrolmentID	FinishTime	Position	ResultStatus	RecordedAt
1	1	00:52:35.0000000	15	Finished	2026-09-07 12:28:21.8433333
2	2	01:48:20.0000000	8	Finished	2026-09-07 12:28:21.8433333
3	3	00:29:45.0000000	5	Finished	2026-09-07 12:28:21.8433333

OUTPUT

EventID	EventName	EventDate	Location	Organiser	Status					
1	Polokwane City Marathon	2026-10-10	Polokwane	John Mokoena	Open					
2	Limpopo Fun Run	2026-11-07	Tzaneen	Maria Molale	Open					
3	Spring Cycling Race	2026-09-20	Mokopane	John Mokoena	Open					
EventName	CategoryName	DistanceKM	EntryFee	MaximumParticipants						
Limpopo Fun Run	5 KM	5.00	50.00	300						
Limpopo Fun Run	10 KM	10.00	100.00	250						
Polokwane City ...	5 KM	5.00	100.00	500						
Polokwane City ...	10 KM	10.00	150.00	500						
Polokwane City ...	21 KM	21.10	250.00	300						
Polokwane City ...	42 KM	42.20	350.00	200						
Spring Cycling ...	10 KM	10.00	200.00	200						
Spring Cycling ...	21 KM	21.10	300.00	150						
EnrolmentID	Participant	EventName	CategoryName	RaceNumber	Status	EnrolmentDate				
1	Thabo Nkosi	Polokwane City Marathon	10 KM	R001	Confirmed	2026-09-07 12:28:21.83333333				
2	Lerato Mabena	Polokwane City Marathon	21 KM	R002	Confirmed	2026-09-07 12:28:21.83333333				
3	Thabo Nkosi	Limpopo Fun Run	5 KM	R003	Confirmed	2026-09-07 12:28:21.83333333				
4	Lerato Mabena	Spring Cycling Race	10 KM	R004	Confirmed	2026-09-07 12:28:21.83333333				

OUTPUT

ResultID	Participant	EventName	CategoryName	FinishTime	Position	ResultStatus					
1	3	Thabo Nkosi	Limpopo Fun Run	5 KM	00:29:45.0000000	5	Finished				
2	2	Lerato Mabena	Polokwane City Marathon	21 KM	01:48:20.0000000	8	Finished				
3	1	Thabo Nkosi	Polokwane City Marathon	10 KM	00:52:35.0000000	15	Finished				